DATA ANALYTICS INTERNSHIP

-InternSpark

(AICTE-Listed Internship Program)

REPORT ON

CUSTOMER BEHAVIOR ANALYSIS

(TASK 1)

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Domain : Data Analytics Intern

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About

This project focuses on analyzing e-commerce customer behavior and purchasing patterns using Python and Google Colab. The analysis was performed using data visualization and customer segmentation techniques to understand customer preferences, spending habits, and business trends.

Introduction

Customer behavior analysis helps businesses understand how customers interact with products and services. In this project, e-commerce customer data was analyzed to identify trends, purchasing patterns, and high-value customers. Various visualizations and statistical methods were used to gain meaningful insights from the dataset.

Objectives

- To analyze customer purchasing behavior
- To identify high-value customers
- To study customer demographics and spending patterns
- To perform data cleaning and preprocessing
- To create visualizations for better understanding of data
- To provide business recommendations based on analysis

Dataset Description

The dataset used in this project contains e-commerce customer information and purchasing details. It includes customer demographics, product-related information, and purchase behavior data.

The dataset was analyzed using:-

- Data cleaning techniques
- Duplicate value removal
- Missing value handling
- Statistical analysis

- Data visualization using graphs and heatmaps

The project was implemented using Python libraries such as Pandas, Matplotlib, and Seaborn in Google Colab.

Tools and Technologies Used:-

- GitHub
- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn

Google Colab Implementation and Analysis Result :-

```
In [18]:
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [22]:
from google.colab import files

uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser s

```
Saving ecommerce_customer_data_large.csv to ecommerce_customer_data_large.c
```

```
In [23]:
import os

os.listdir('/content')
```

```
Out[23]: ['.config', 'drive', 'ecommerce_customer_data_large.csv', 'sample_data']
```

In [24]:
```
import pandas as pd

df = pd.read_csv('/content/ecommerce_customer_data_large.csv')

df.head()
```

Out[24]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0

In [25]:
```
df.shape
```

Out[25]: (250000, 13)

In [26]:
```
df.columns
```

Out[26]:
```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```

In [27]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
 #   Column                 Non-Null Count   Dtype
---  ------                 --------------   -----
 0   Customer ID            250000 non-null  int64
 1   Purchase Date          250000 non-null  object
 2   Product Category       250000 non-null  object
 3   Product Price          250000 non-null  int64
 4   Quantity               250000 non-null  int64
 5   Total Purchase Amount  250000 non-null  int64
 6   Payment Method         250000 non-null  object
 7   Customer Age           250000 non-null  int64
 8   Returns                202618 non-null  float64
 9   Customer Name          250000 non-null  object
 10  Age                    250000 non-null  int64
 11  Gender                 250000 non-null  object
 12  Churn                  250000 non-null  int64
dtypes: float64(1), int64(7), object(5)
memory usage: 24.8+ MB
```

In [28]:
```
df.isnull().sum()
```

Out[28]:

	0
Customer ID	0
Purchase Date	0
Product Category	0
Product Price	0
Quantity	0
Total Purchase Amount	0

Total Purchase Amount	0
Payment Method	0
Customer Age	0
Returns	47382
Customer Name	0
Age	0
Gender	0
Churn	0

dtype: int64

In [30]:
```
df = df.dropna()
```

In [31]:
```
df.describe()
```

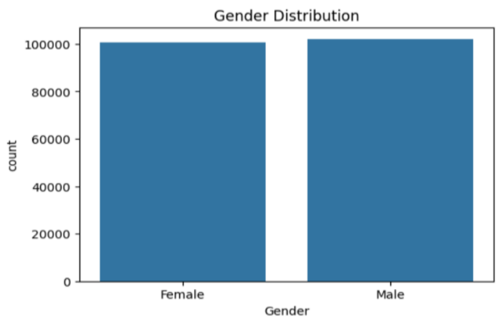
Out[31]:

	Customer ID	Product Price	Quantity	Total Purchase Amount	Customer Age	Returns	Age	Churn
count	202618.000000	202618.000000	202618.000000	202618.000000	202618.000000	202618.000000	202618.000000	202618.000000
mean	25020.140234	254.899066	3.008420	2725.817662	43.817923	0.500824	43.817923	0.201088
std	14412.388674	141.720425	1.414339	1442.268491	15.356067	0.500001	15.356067	0.400815
min	1.000000	10.000000	1.000000	100.000000	18.000000	0.000000	18.000000	0.000000
25%	12599.250000	133.000000	2.000000	1478.000000	30.000000	0.000000	30.000000	0.000000
50%	25018.000000	255.000000	3.000000	2727.000000	44.000000	1.000000	44.000000	0.000000
75%	37444.000000	377.000000	4.000000	3975.000000	57.000000	1.000000	57.000000	0.000000
max	50000.000000	500.000000	5.000000	5350.000000	70.000000	1.000000	70.000000	1.000000

In [32]:
```
plt.figure(figsize=(6,4))
sns.countplot(x='Gender', data=df)

plt.title("Gender Distribution")

plt.show()
```

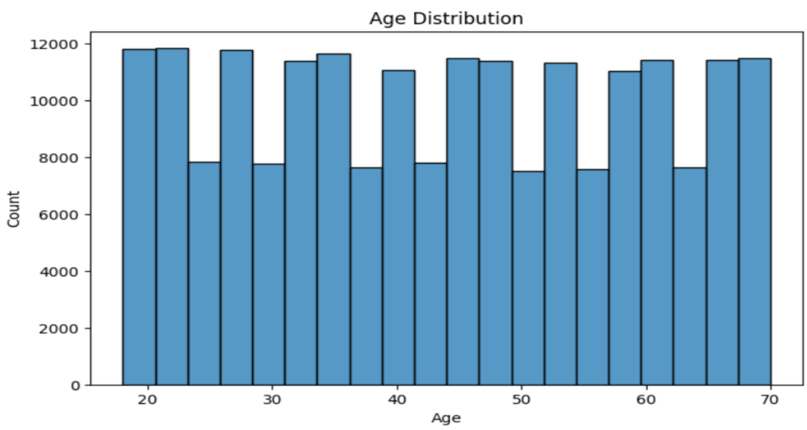


In [33]:
```
plt.figure(figsize=(8,5))

sns.histplot(df['Age'], bins=20)

plt.title("Age Distribution")

plt.show()
```



In [34]:
```python
df.duplicated().sum()
```
Out[34]:
```
np.int64(0)
```
In [36]:
```python
df = df.drop_duplicates()
```
In [37]:
```python
df.columns
```
Out[37]:
```
Index(['Customer ID', 'Purchase Date', 'Product Category', 'Product Price',
       'Quantity', 'Total Purchase Amount', 'Payment Method', 'Customer Age',
       'Returns', 'Customer Name', 'Age', 'Gender', 'Churn'],
      dtype='object')
```
In [38]:
```python
df.head(10)
```
Out[38]:

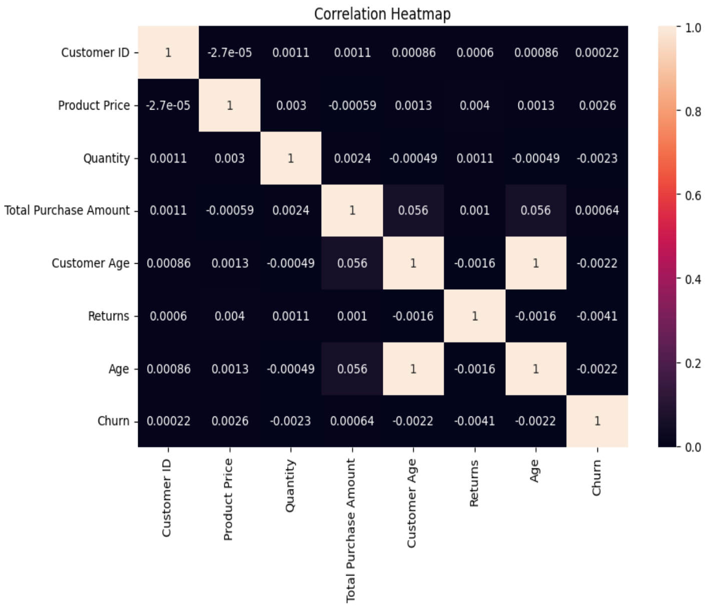
	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0
5	13738	2022-08-25	Home	191	3	3722	Credit Card	27	1.0	Lauren	27	Female	0
7	13738	2023-02-05 19:31:48	Books	370	5	1486	Cash	27	1.0	Lauren Johnson	27	Female	0
9	13738	2023-02-09 00:53:14	Electronics	40	4	4327	Cash	27	0.0	Lauren Johnson	27	Female	0
11	33969	2023-01-05 11:15:27	Home	304	1	3883	PayPal	27	1.0	Carol Allen	27	Male	0
12	33969	2023-07-18 23:36:50	Books	54	2	4187	PayPal	27	0.0	Carol Allen	27	Male	0

In [39]:
```python
plt.figure(figsize=(10,6))

sns.heatmap(df.corr(numeric_only=True), annot=True)

plt.title("Correlation Heatmap")

plt.show()
```



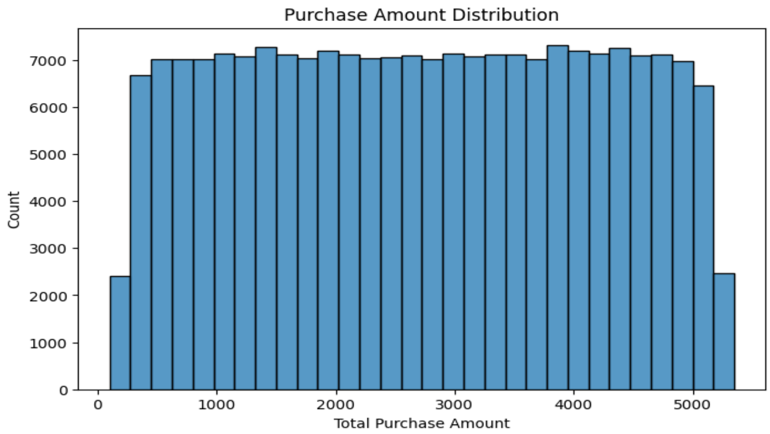
In [41]:

```
plt.figure(figsize=(8,5))

sns.histplot(df['Total Purchase Amount'], bins=30)

plt.title("Purchase Amount Distribution")

plt.show()
```



In [43]:

```
high_value = df[df['Total Purchase Amount'] > 500]

high_value.head()
```

Out[43]:

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	44605	2023-05-03 21:30:02	Home	177	1	2427	PayPal	31	1.0	John Rivera	31	Female	0
1	44605	2021-05-16 13:57:44	Electronics	174	3	2448	PayPal	31	1.0	John Rivera	31	Female	0
2	44605	2020-07-13 06:16:57	Books	413	1	2345	Credit Card	31	1.0	John Rivera	31	Female	0
3	44605	2023-01-17 13:14:36	Electronics	396	3	937	Cash	31	0.0	John Rivera	31	Female	0
4	44605	2021-05-01 11:29:27	Books	259	4	2598	PayPal	31	1.0	John Rivera	31	Female	0

In [47]:

```
plt.figure(figsize=(8,5))

sns.countplot(x='Product Price', data=df)

plt.xticks(rotation=90)

plt.title("Product Price Purchase")

plt.show()
```

In [47]:

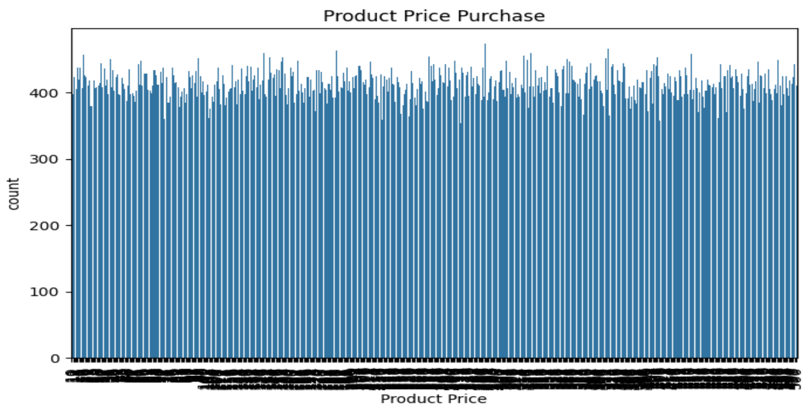
```
plt.figure(figsize=(8,5))

sns.countplot(x='Product Price', data=df)

plt.xticks(rotation=90)

plt.title("Product Price Purchase")

plt.show()
```



Findings:-

1. High spending customers contribute most revenue.

2. Some product categories are purchased more frequently.

3. Customer behavior varies across age and gender.

4. Purchase trends show specific customer preferences.

5. Cleaned dataset improved analysis accuracy.

Recommendations:-

1. Provide special offers to high-value customers.

2. Introduce loyalty reward programs.

3. Focus marketing on popular product categories.

4. Send personalized product recommendations.

5. Improve customer retention through discounts and engagement campaigns.

Code Link :-

https://github.com/Shadnmnn34/InternSpark--InternShip/blob/d83c048f34c8bd5d1e2ff4cf41fdf6a9d679c235/Customer%20behaviour%20analysis/CusomerBehaviourAnalysis.ipynb